Activity trace — accelerating matched SHD learning

ar073 · 19 July 2026 · Draft

Checkpoint CP-001

The next exploratory night is registered before any exp070 training result. This checkpoint has immutable private source SHA-256 prefix `5189910c09c9`. Its checkpoint time is `2026-07-19 06:49:09.368 UTC`, and it has no predecessor within this night. The raw snapshot is held read-only outside the repository. Hidden reasoning, tool payloads, injected environment material, credentials, private paths, addresses, and infrastructure identifiers are excluded. The visible attachment path is explicitly redacted; the request itself is reproduced verbatim.

Decision and action ledger

1. Pull request 52 is open and draft on the requested existing branch. Main is unchanged.
2. Exp069′s complete epoch histories, firing diagnostics, gradients, rasters, split provenance, billing, and official-test prohibition were inspected.
3. A development-data-only audit rejected observation-window truncation as the likely bottleneck. The first registered candidate changes only the matched temporal resolution from 1 ms to 2 ms.
4. The ordered ladder, first-passing selection rule, promotion threshold, forty-epoch success criterion, kill criteria, one-seed policy, locked model comparison, and fresh paid-compute gate are fixed in ar072.
5. No exp070 runner, training result, cloud job, official-test access, or spend exists at this checkpoint. Pending work is the candidate-1 implementation, focused checks, local smoke, cost estimate, and paid-compute authority gate.

Privacy-sanitized visible transcript

User 2026-07-19 06:44:12.361 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Files mentioned by the user:

/goal Design and execute the next exploratory SHD experiment on PR #52, aimed a…: (PRIVATE ATTACHMENT PATH REDACTED)

The attached pasted text file(s) contain the user's request. Read and act on that content.

My request for Codex:

Attached request

/goal Design and execute the next exploratory SHD experiment on PR #52, aimed at materially improving the matched COBA/PING baseline while preserving a fair scientific comparison.

Context:
- Continue from exp069 on branch night/spiking-heidelberg-digits/ar070.
- exp069 trained for 80 epochs and reached 43.87% held-out validation accuracy for COBA and 48.04% for PING.
- The additional epochs provided insufficient benefit. Use much shorter runs while developing, then return to a 40-epoch matched run after selecting a convincing configuration.
- One seed is sufficient for this exploratory phase. Do not introduce multiple seeds until there is a claim worth defending.
- Never access or tune against the official SHD test set during exploration.

Scientific objective:
Identify a mechanistically justified change that substantially improves SHD learning—ideally improving both cells while revealing whether PING offers an advantage over COBA.

Investigate likely bottlenecks in this order:
1. input temporal representation/resolution and shared input scaling;
2. initialization and optimization settings;
3. inhibitory recruitment, firing-rate balance, and gradient flow;
4. existing or carefully added neuron/synapse time-constant mechanisms;
5. only if justified by the diagnostics, a new matched readout structure.

Locked comparison:
- Preserve Dale constraints.
- Keep model capacity, hidden population sizes, E/I counts, connectivity, and recurrence structure fixed.
- COBA uses no voltage-gradient dampening.
- PING uses voltage-gradient dampening 1000.
- Keep dataset split, checkpoint-selection rule, batching, and all non-cell-specific settings matched.
- Any new mechanism, including a readout change, must be applied identically to COBA and PING unless the difference is intrinsic to their registered cell definitions.
- Do not change the scientific design after seeing results without recording the result and obtaining approval for the revised design.

Readout and tools/snn authority:
- Do not casually search over losses or readouts.
- Begin with exp069's existing readout and loss.
- A new readout structure is allowed only when the diagnostics provide a clear scientific justification and it is pre-registered before its results are observed.
- Keep the classification objective matched and avoid introducing a readout that obscures the recurrent-

cell comparison.
- You are authorized to make careful edits or additions to tools/snn when needed, including support for a justified new readout structure or neuron/synapse mechanism.
- Preserve backward compatibility, keep changes narrowly scoped, add focused tests, and document the engine commit anywhere it affects scientific interpretation.
- Stop and ask if the required engine change becomes broad, invasive, or changes the meaning of the COBA/PING comparison.

Protocol:
1. Inspect exp069's implementation, artifacts, learning curves, firing diagnostics, rasters, and epoch-by-epoch behavior. Diagnose the most likely bottleneck before changing anything.
2. Create and lock a timestamped mandate in the next arNNN article. Pre-register the hypothesis, allowed intervention sequence, selection rule, success criterion, and kill criterion.
3. Run a one-seed local smoke stage first.
4. Iterate using very short registered runs, preferably about 5 epochs and no more than 10 unless justified. Use only the held-out validation split.
5. Use a small sequential set of scientifically motivated interventions, normally changing one factor at a time. Do not conduct an unconstrained hyperparameter sweep.
6. Record every meaningful attempt, including null, failed, and killed attempts, before proceeding.
7. Promote a configuration only if its short-run curves, losses, gradients, and firing diagnostics provide credible evidence that it should outperform the exp069 trajectory.
8. Freeze the promoted configuration and run matched COBA and PING training for 40 epochs.
9. Compare the new run against exp069 at matched epochs and against exp069's final 80-epoch checkpoint. Do not claim improvement through selective checkpoint reporting.
10. Do not use the official SHD test set. Stop at the held-out-validation human-review gate.

Compute:
- Run focused local checks first.
- Estimate cloud runtime and cost before requesting paid-compute authorization.
- Do not assume an earlier RunPod budget applies to this experiment.
- If authorized, use at most one pod per cell, respect the approved ceiling, and reap every pod immediately.
- Do not run the full test suite on the 4 GB Hetzner host. Use focused lightweight tests and demolab build.

Deliverables:
- Implement the experiment through experiments/expNNN.py and narrowly scoped tools/snn changes where justified.
- Produce artifacts/data/expNNN with complete provenance, numbers.json, reproducer, learning curves, firing-rate and gradient diagnostics, and matched input/E/I rasters.
- Create a cold-readable writings/expNNN.typ whose numbers and figures are sourced from recorded artifacts.
- Maintain timestamped scientific records in the relevant arNNN article: decisions, hypotheses, actions, failures, anomalies, costs, commits, results, conclusions, and pending work.
- Continue the existing research activity-log article with meaningful timestamped, hash-linked checkpoints.

- Make focused commits for implementation, engine changes, and every meaningful scientific attempt, including killed attempts.
- Build and validate the Demolab publication.
- Push the existing branch and update PR #52 with useful milestone comments.
- Do not merge the PR or modify main.
- Finish at the human-review gate with a concise evidence summary, exact compute spend, validation performed, rendered article/report links, relevant commit links, and the PR link.

Assistant 2026-07-19 06:44:16.949 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll load the lab runbook and the attached `/goal`, then verify the current branch/PR and resume from the recorded exp069 state. I'll also use the GitHub publishing workflow when the experiment reaches a commit/push milestone.

Assistant 2026-07-19 06:44:20.058 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The direct `demolab` command is not installed on this host, so I'm using the repository-prescribed `uvx demolab-cli docs` fallback before touching the experiment.

Assistant 2026-07-19 06:44:24.580 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

This is an AUTORESEARCH continuation, so I'm loading that runbook and its programme rules, plus the attached mandate and GitHub commit/push instructions. I won't start paid compute until I've reconciled the registered design with the current branch and its existing records.

Assistant 2026-07-19 06:45:10.455 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The programme is in "changes requested" state: you have explicitly approved a new exploratory night on the existing PR branch. I'm treating the attached goal as the mandate-gate approval, preserving its scientific terms verbatim, and keeping `main` untouched as requested; the branch commit will remain the pre-result anchor.

Assistant 2026-07-19 06:45:44.949 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The exp069 evidence is intact locally, including full epoch histories and rasters. I'm now quantifying its early-epoch bottleneck, checking the next free IDs, and locating the existing private-trace custody pattern so the new night begins with a complete, sanitized checkpoint rather than a reconstructed summary.

Assistant 2026-07-19 06:47:51.951 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first input audit is testing two concrete failure modes on development data only: how much speech exp069 discarded after 1.0 seconds, and how much temporal information coarser bins would collapse. This precedes intervention selection and does not touch the official test split.

Assistant 2026-07-19 06:49:07.003 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The diagnosis points to temporal discretization as the first clean intervention. Exp069 discarded only 0.017% of development events after 1 second, so extending duration is unlikely to move accuracy. A 2 ms bin collapses only 0.094% of events in a deterministic 1/8 development sample, while halving backpropagation depth; 4 ms already collapses 2.04% and is coarse relative to the 2 ms AMPA time constant. I'm preregistering 2 ms first, before observing any new training result.

Checkpoint CP-002

Candidate 1 is implemented and its local smoke gate passes. This checkpoint follows mandate anchor `cac3d07`. Its immutable private source has SHA-256 prefix `8b7c5612397e`, and its checkpoint time is `2026-07-19 07:04:04.503 UTC`.

Decision and action ledger

1. The exp070 runner changes only temporal resolution and short-run duration while reusing exp069′s split, checkpoint selection, diagnostics, rasters, and cloud orchestration. `tools/snn` is unchanged.
2. A pre-smoke audit found that inherited backup/restore plumbing could copy an existing official-test cache even though it never evaluated it. Exp070 now redirects the engine to a development-only alias, closing that protocol ambiguity before training.
3. Focused Ruff, `ty`, compilation, and two-pod dry-run checks pass. The dry run created no pod.
4. Both 2 ms smoke cells are finite and active, use the exact exp069 split hashes, and report no skipped or non-finite updates. Spend remains 0 USD.
5. Pending work is a commit/push checkpoint and fresh authority for the matched five-epoch cloud pair. No paid compute may start before that approval.

Visible messages added in CP-002

Assistant 2026-07-19 06:50:08.850 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I’ve frozen the first private transcript checkpoint outside the repository as read-only, with SHA-256 prefix `5189910c09c9`. The public trace will redact the attachment’s private path and preserve the message plus attached mandate verbatim; no credentials, tokens, addresses, environment values, or infrastructure identifiers will enter the article.

Assistant 2026-07-19 06:52:40.014 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate and CP-001 trace are written. I’m validating three things before the checkpoint commit: the trace JSON is syntactically sound, the publishable files contain no sensitive path/token/address patterns, and the entire Demolab collection renders with the new articles.

Assistant 2026-07-19 06:54:07.371 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The build succeeds with 40 entries. It also refreshed every tracked PDF and emitted several stale untracked stub PDFs, which is normal build churn rather than experiment content. I’m narrowing the checkpoint to the two new article PDFs, the updated book, the articles, and the sanitized trace data; existing articles/PDFs remain untouched.

Assistant 2026-07-19 06:55:56.532 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

CP-001 passes the full 40-entry build and the fail-closed sensitive-pattern scan. I’m committing only the new mandate, activity trace/data, their PDFs, and the updated book, then pushing the existing branch so PR #52’s preview advances.

Assistant 2026-07-19 06:57:23.024 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first candidate needs no engine change: exp069 already exposes timestep, preserves the exact split, and records the required losses, gradients, rates, and checkpoints. I’m implementing exp070 as a narrow orchestration layer over that validated runner, with 2 ms hardcoded, a local smoke default, two short cloud cells only, and an automatic promotion/kill decision against the registered epoch-five baselines.

Assistant 2026-07-19 06:59:38.280 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The focused checks exposed only two type-level literal assignments, now fixed. Before smoke, I caught a stricter protocol issue in the inherited plumbing: exp069 temporarily backed up any existing `(OFFICIAL

TEST CACHE PATH REDACTED)` while installing validation under the CLI's fixed filename. It never evaluated that file, but exp070's "no access" rule is stronger. I'm isolating the trainer's SHD alias in an experiment-specific directory so the official test file is neither copied nor opened.

Assistant 2026-07-19 07:00:27.614 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The isolated runner now passes Ruff, `ty`, compilation, and a two-pod dry run; the dry run created nothing. I'm starting the registered local 128/128, two-epoch candidate-1 smoke. It will read only official training data to derive development train/validation subsets, and the engine will see those subsets in the exp070-specific alias directory.

Assistant 2026-07-19 07:01:42.047 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Candidate 1's COBA smoke completed cleanly at 2 ms: finite two-epoch training, active 16 Hz excitatory firing, zero inhibitory activity as registered, and no update failures. PING is now running under dampening 1000 in the same isolated split.

Assistant 2026-07-19 07:04:01.063 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered local smoke passed both cells. COBA finished active at 16.18 Hz E; PING at 4.27/15.00 Hz E/I; both had finite losses, exact exp069 split hashes, and no validation/test alias leakage. I'm promoting the smoke evidence into the committed artifact record, then checkpointing the runner and timestamped trace before asking for paid-compute authority.

Checkpoint CP-003

The cloud boundary now fails closed on scientific and billing integrity. This checkpoint follows smoke commit `135b82e`. Its immutable private source has SHA-256 prefix `20a7b2d23008`, and its checkpoint time is `2026-07-19 07:11:32.044 UTC`.

Decision and action ledger

1. The expected first-candidate plus promoted-final path is approximately 2 USD. The requested hard ceiling is 10 USD across the entire registered ladder, with one pod per cell and two concurrent pods maximum. Authority has not yet been granted, so no dispatch occurred.
2. GitHub Actions reports the same ten repository-wide `ty` diagnostics in unchanged older experiments and tests. Exp070 produces no CI diagnostic, and those unrelated files remain out of scope.
3. Two focused tests pin the first-passing promotion gate: both cells must clear the accuracy and loss thresholds, and non-finite or saturated activity kills the candidate.

4. Artifact publication now refuses to proceed without finite authoritative provider spend and confirmation that zero pods remain. Ruff, focused `ty`, compilation, both tests, and the missing-ledger failure check pass.
5. Pending work remains fresh paid-compute authorization, the matched five-epoch pair, exact billing reconciliation, and the registered adaptive decision. Spend remains 0 USD.

Visible messages added in CP-003

Assistant 2026-07-19 07:07:24.169 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The local checkpoint is clean: focused Ruff, \`ty\`, compilation, JSON validation, sensitive-pattern scan, exact split/config assertions, and the live 40-entry Demolab rebuild all pass. I'm committing this smoke attempt now; \`tools/snn\` remains unchanged and RunPod spend is still exactly $0.

Assistant 2026-07-19 07:08:15.643 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Candidate 1 is ready for cloud testing.

Estimated RunPod cost:

- Five-epoch 2 ms pair: about $0.40 expected
- If promoted, forty-epoch 2 ms pair: about $1.50 expected
- Expected combined path: about $2
- Proposed hard ceiling across all registered candidates and final run: $10
- Maximum concurrency: two pods, exactly one per cell
- Backstops: one hour for short runs, two hours for final runs
- All pods reaped immediately

Current spend is $0. Smoke checkpoint: \`135b82e\`; PR #52 is updated.

Please authorize RunPod spending up to the $10 hard ceiling for exp070.

Assistant 2026-07-19 07:08:32.804 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Paid compute remains gated pending your explicit authorization, so I won't dispatch anything. I'll use this continuation to harden exp070's remaining registered paths, verify the pushed preview/checks, and ensure cost reconciliation and the eventual 40-epoch promotion cannot proceed with missing evidence.

Assistant 2026-07-19 07:08:46.471 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR #52 is still draft and unmerged; its preview is updating. The new lint check is red, so I'm using the GitHub CI-diagnosis workflow to verify whether exp070 introduced anything or whether this is the already-documented repository-wide type debt. I'll only change files in scope if the logs point to exp070.

Assistant 2026-07-19 07:09:26.515 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

CI confirms exp070 adds no diagnostic: the lint failure is the same ten pre-existing `ty` errors in unchanged exp041/048/054/065 and their older tests. I'm not expanding scope to fix them. One in-scope hardening remains before dispatch: make short-result publication fail closed unless exact provider billing and zero active pods are recorded, and enforce the registered validation-rate saturation checks explicitly.

Assistant 2026-07-19 07:11:19.187 UTC

Role: assistant · Checkpoint: CP-003 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The hardening checks now pass: two focused promotion-gate tests verify that both cells must clear accuracy/loss criteria and that non-finite or saturated activity kills the candidate. Publication also fails immediately if the exact billing ledger or zero-pod confirmation is absent. I'm recording this CI/provenance checkpoint while the paid-compute gate remains closed.

Checkpoint CP-004

Fresh paid-compute authority is explicit and bounded. This checkpoint follows hardening commit `2bc22d3`. Its immutable private source has SHA-256 prefix `c20363a95684`, and its checkpoint time is `2026-07-19 07:21:26.366 UTC`.

Decision and action ledger

1. The scientist authorized a 10 USD hard total ceiling across exp070, with at most two concurrent pods and exactly one cell per pod.
2. Short candidates carry one-hour pod backstops. A promoted forty-epoch pair may use two-hour backstops. Every pod must be reaped immediately.
3. Before this checkpoint the fleet was empty, spend was 0 USD, the branch was clean at `2bc22d3`, and the pull request remained draft and unmerged.
4. The next action is limited to the registered five-epoch 2 ms pair. No later candidate or final pair is dispatched before applying the registered result-dependent decision.

Visible messages added in CP-004

Assistant 2026-07-19 07:14:17.507 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Exp070 is fully hardened and ready to dispatch from `2bc22d3`; current spend is $0.

Requested authorization: RunPod up to a $10 hard total ceiling, maximum two concurrent pods (one COBA, one PING), with one-hour short-run and two-hour final-run backstops. Expected spend is about $2 if candidate 1 is promoted.

Please explicitly authorize this paid compute.

Assistant 2026-07-19 07:14:28.941 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The only remaining gate is unchanged: paid RunPod authority has not been granted, so dispatching would violate the registered mandate. I'm doing a final safety check for zero active pods, clean remote state, and the latest PR preview/check status; if those are intact, I must formally stop at this authority blocker rather than spend without permission.

Assistant 2026-07-19 07:14:44.983 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Blocked only on fresh paid-compute authorization.

Safety state:

- Active pods: 0
- Spend: $0
- Branch clean and pushed at `2bc22d3`
- PR #52 remains draft and unmerged
- Candidate-1 smoke and focused checks pass
- CI lint failure remains unrelated pre-existing type debt

Authorize RunPod up to the $10 hard ceiling to resume.

User 2026-07-19 07:21:02.887 UTC

Role: user · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

authorized

Assistant 2026-07-19 07:21:09.728 UTC

Role: assistant · Checkpoint: CP-004 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Authorization received for the proposed exp070 envelope: $10 hard total ceiling, at most two concurrent pods with one per cell, one-hour short-run backstops and two-hour final-run backstops, with immediate reaping. I'll checkpoint this authority first, verify the fleet is empty, then dispatch only the registered five-epoch 2 ms COBA/PING pair from `2bc22d3`.

Checkpoint CP-005

Only the registered candidate-1 pair is dispatched. This checkpoint follows authority commit `cd82023`. Its immutable private source has SHA-256 prefix `3b4b22492a5c`, and its checkpoint time is `2026-07-19 07:23:43.025 UTC`.

Decision and action ledger

1. The dry run found zero existing pods and exactly two jobs: one COBA and one PING cell at 2 ms for five epochs.
2. Two 5090 pods were created at 0.99 USD/hour each, pinned to `cd82023`, with one-hour self-removal backstops. The post-dispatch fleet count was exactly two, so worst-case exposure for this attempt is 1.98 USD.
3. Pod identifiers and other infrastructure details remain only in the private trace. The public record retains names, counts, rates, commit, and bounds.
4. Pending work is self-termination, collection, exact provider billing, artifact publication, and the preregistered promotion decision.

Visible messages added in CP-005

Assistant 2026-07-19 07:22:57.584 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The authority checkpoint is pushed at `cd82023`; the scientific runner is unchanged from the tested `2bc22d3` state. I'm performing the final dry run and fleet check now. If it still resolves to exactly two jobs and zero existing pods, I'll dispatch the five-epoch pair under the one-hour backstops.

Assistant 2026-07-19 07:23:13.646 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The dry run is exact: zero existing pods, then one COBA and one PING job on two 5090 pods, with no extra work. Worst-case exposure for this short pair is $1.98, below the $10 cumulative ceiling. Dispatching now from `cd82023`; I'll verify the fleet count immediately afterward.

Assistant 2026-07-19 07:23:36.032 UTC

Role: assistant · Checkpoint: CP-005 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The registered pair is live: exactly two pods at $0.99/hour each, one named COBA and one PING, pinned to `cd82023`; no third pod exists. The one-hour guards cap this attempt at $1.98. I'm checkpointing the dispatch without publishing pod identifiers, then monitoring for self-termination before collection and billing reconciliation.

Checkpoint CP-006

Candidate 1 is killed after the registered five-epoch comparison. This checkpoint follows dispatch commit `1bba764`. Its immutable private source has SHA-256 prefix `cf6b7ebf703b`, and its checkpoint time is `2026-07-19 07:34:41.277 UTC`.

Decision and action ledger

1. Both pods self-terminated about eight minutes after dispatch; the active fleet is zero and the complete cell outputs were collected.
2. COBA reached 23.53% validation accuracy at epoch five, 4.41 percentage points below exp069, while cross-entropy worsened by 0.206 to 2.678.
3. PING reached 27.70%, 4.90 points below exp069, while cross-entropy worsened by 0.148 to 2.592.
4. Both cells were finite and active with no skipped or non-finite updates, so this is a scientific kill rather than a plumbing failure. The 2 ms change fails every registered promotion requirement.
5. Exact provider billing has not yet appeared in the billing history. The publication gate remains closed until the authoritative spend is recorded; no estimate is substituted. Candidate 2 has not been implemented or run.

Visible messages added in CP-006

Assistant 2026-07-19 07:25:21.662 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Dispatch checkpoint `1bba764` is pushed; I also corrected the PR comment's literal commit placeholder without editing history. Both jobs remain active. I'm switching to a single bounded monitor that reports only timestamp, job names, and fleet count; it will exit as soon as both pods self-terminate.

Assistant 2026-07-19 07:26:21.152 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both candidate jobs remain active at 07:25:59 UTC. Fleet count is still exactly two, estimated exposure is under $0.10, and the one-hour safeguards remain intact. No duplicate process or additional pod has been created.

Assistant 2026-07-19 07:28:07.372 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The pair is still healthy and bounded at 07:27:32 UTC. This is consistent with cold image startup plus staging before the five short epochs; estimated combined exposure is about $0.15. I'm continuing the same monitor.

Assistant 2026-07-19 07:30:33.592 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both pods remain present at 07:30:07 UTC, about seven minutes after dispatch. Estimated combined exposure is about $0.23. That remains within the expected startup-plus-short-run window and far below the $1.98 attempt cap.

Assistant 2026-07-19 07:31:15.947 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Both training pods self-terminated by 07:31:10 UTC, about eight minutes after dispatch; active fleet is now zero. I'm collecting the shared-volume outputs with the bounded collector, then I'll verify both five-epoch records and reconcile exact provider billing before allowing publication or any next candidate.

Assistant 2026-07-19 07:33:04.359 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The first registered cloud candidate has finished and is decisively killed: moving to 2 ms reduced held-out accuracy and worsened cross-entropy in both COBA and PING. I'm reconciling exact provider billing now, then I'll publish and checkpoint this failed attempt before touching candidate 2.

Assistant 2026-07-19 07:34:22.996 UTC

Role: assistant · Checkpoint: CP-006 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Provider billing is lagging behind pod termination, so I'm not inventing an "exact" cost. The fleet is confirmed empty; while the authoritative entries settle, I'm preserving the candidate's outputs and preparing the timestamped killed-attempt record.

Checkpoint CP-007

Candidate 2 is implemented and passes its local smoke gate. This checkpoint follows the killed-attempt commit `9e1385a`. Its immutable private source has SHA-256 prefix `b12370990974`, and its checkpoint time is `2026-07-19 07:47:15.296 UTC`.

Decision and action ledger

1. Candidate 2 restores 1 ms and changes only the matched shared input-weight mean from 0.9 to 1.2. Candidate 1′s outputs remain separately addressable.
2. Focused Ruff, type, compile, and promotion-gate checks pass. `tools/snn` is unchanged.
3. The required 128-training/128-validation two-epoch smoke passes in both cells. COBA ends at 26.79 Hz E; PING ends at 3.24 Hz E and 17.00 Hz I. Losses and gradients are finite and no update is skipped.
4. The exact registered split hashes are retained and the development-only alias contains no official-test file. The two-job cloud dry run creates no pod and resolves to exactly one job per cell.

5. Candidate-1 provider billing is still delayed. The fleet is empty and candidate 2 has not been dispatched at this checkpoint.

Visible messages added in CP-007

Assistant 2026-07-19 07:37:37.732 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The killed candidate is now committed and pushed at `9e1385a`, with its timestamped activity checkpoint and PR milestone comment. No later candidate has started. I'm waiting only for the provider's authoritative billing rows so this attempt can be published with exact spend, then I'll advance to the preregistered 1.2 input-scale smoke.

Assistant 2026-07-19 07:41:19.944 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Candidate 2 is now implemented exactly as registered: 1 ms is restored and only the shared input-weight mean changes from 0.9 to 1.2. The runner can still reproduce and publish the killed 2 ms attempt without mixing its files. Focused lint, type, compile, and promotion-gate tests pass; I'm running the required local 128/128 two-epoch smoke now.

Assistant 2026-07-19 07:46:43.509 UTC

Role: assistant · Checkpoint: CP-007 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The candidate-2 smoke passes: both cells are finite and active, use the exact registered 7,340/816 split hashes, and the development-only alias contains no official-test file. COBA ends at 26.79 Hz E; PING at 3.24 Hz E and 17.00 Hz I, with no skipped/non-finite updates. I'm checkpointing this smoke and verifying the two-job dry run before any paid dispatch.

Checkpoint CP-008

Only the registered candidate-2 pair is dispatched. This checkpoint follows smoke commit `04d5a28`. Its immutable private source has SHA-256 prefix `a0717f4a44f2`, and its checkpoint time is `2026-07-19 07:50:04.550 UTC`.

Decision and action ledger

1. The clean dry run found zero existing pods and exactly two candidate-2 jobs: one COBA and one PING cell at 1 ms with shared input mean 1.2.
2. Two 5090 pods were created at 0.99 USD/hour each, pinned to `04d5a28`, with one-hour self-removal backstops. The verified post-dispatch fleet count is exactly two.
3. Candidate 1′s exact billing is still delayed, but its hard bound is 1.98 USD. The two short attempts therefore have a cumulative hard bound of 3.96 USD, below the authorized 10 USD ceiling.
4. Infrastructure identifiers remain only in the immutable private trace. Pending work is self-termination, collection, provider reconciliation, and the locked promotion decision.

Visible messages added in CP-008

Assistant 2026-07-19 07:49:25.308 UTC

Role: assistant · Checkpoint: CP-008 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Candidate 2 is committed and pushed at `04d5a28`. Exact candidate-1 billing is still absent from the provider history, but its one-hour guard makes the hard cumulative exposure at most $1.98; adding this registered short pair can raise that bound only to $3.96, safely below the authorized $10 ceiling. I'm posting the smoke milestone, rerunning the clean dry run, then dispatching exactly one pod per cell.

Assistant 2026-07-19 07:49:57.899 UTC

Role: assistant · Checkpoint: CP-008 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The candidate-2 pair is live from clean commit `04d5a28`: exactly two 5090 pods at $0.99/hour, one COBA and one PING, with no third pod. The one-hour safeguards put the cumulative worst-case bound at $3.96 across both short attempts. I'm checkpointing this dispatch without publishing infrastructure identifiers, then monitoring until the fleet returns to zero.